1 Case Studies

We present two evaluation rollouts of SLIC-RANDOM at the 65 nm technology node, covering different topologies, initial operating conditions, and target specifications. Case 1 uses a fully-differential folded-cascode amplifier from the in-distribution split and starts from a valid design that meets only the bandwidth target. Case 2 uses a held-out fully-differential current-mirror amplifier from the out-of-distribution split and starts from an invalid design. Both reach all four targets in four turns, including recovery from an invalid intermediate design. Model responses are reproduced as generated, including their intermediate interpretations. Long graph descriptions and repeated device features are omitted where indicated.

1.1 Case 1: In-Distribution Folded-Cascode Amplifier

Figure 1 shows the folded-cascode amplifier used in this case. This topology contains 20 transistors across the amplifier core and three bias-generator blocks, with 31 tunable parameters. The initial gain is 36.7 dB below the target, power is about 34 times the budget, and phase margin is outside the required window. The policy first recovers from an invalid update, then increases bandwidth while retaining sufficient gain and acceptable power.

Task Setup

Topology	Fully-differential folded-cascode amplifier
Evaluation split	In-distribution
Technology	65 nm CMOS (PTM), $V_{DD} = 1.2$ V
Gain target	≥ 61.7 dB
Power target	$\leq 425.7\ \mu$W
Bandwidth target	≥ 43.1 kHz (-3 dB bandwidth)
Phase-margin target	$30^\circ \leq \text{PM} \leq 90^\circ$
Evaluation tolerance	5%, as defined in the main evaluation
Initial gain	25.0 dB
Initial power	14,349 μW
Initial bandwidth	111 kHz
Initial phase margin	-36°
Initial in-band peaking	0.0 dB
Initial validity	Valid. Only the bandwidth target is met.

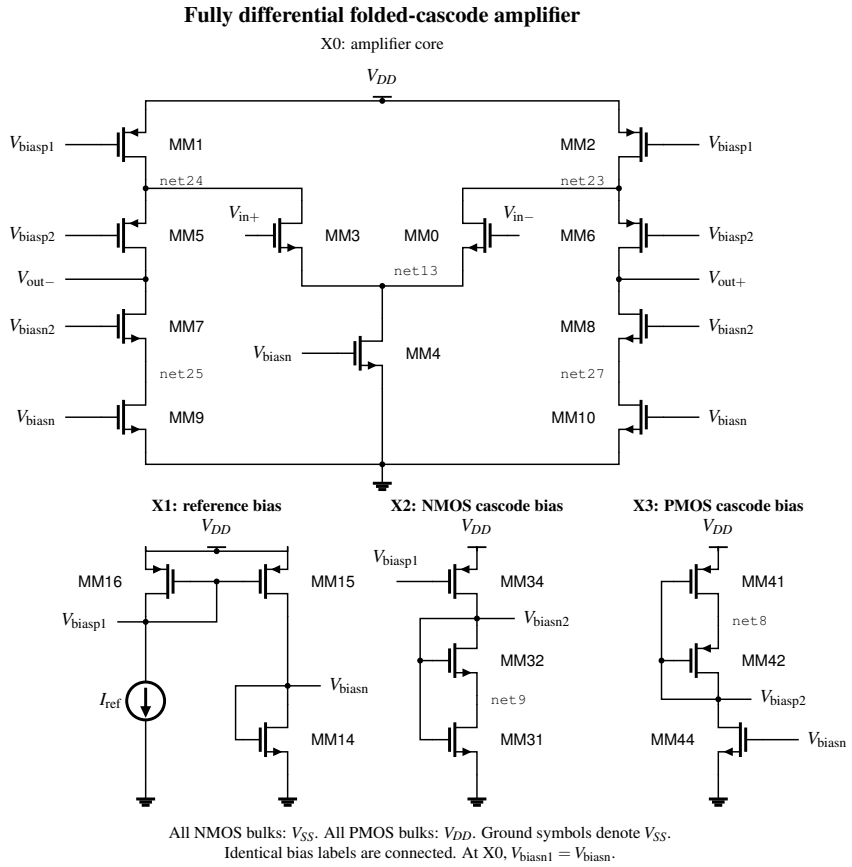


Figure 1: Fully differential folded-cascode amplifier used in Case 1. X0 is the amplifier core and X1–X3 generate its bias voltages. Device identifiers and connections follow the supplied netlist. In the top-level instance, V_{biasn1} is tied to V_{biasn}. Identical bias labels denote the same electrical net.

Structured System Input Excerpt

```
<GRAPH_INFO>
TOPOLOGY_NAME: Fully_differential_folded_cascode
</GRAPH_INFO>
<GRAPH_NODE_IDMAP>
type=pin: @0=vdd @1=vss
type=current_source: @70=X1.I0
type=n_mosfet: @10=X0.MM0 @114=X3.MM44 @16=X0.MM10 @30=X0.MM3 @35=X0.MM4 @50=X0.MM7
  @55=X0.MM8 @60=X0.MM9 @73=X1.MM14 @88=X2.MM31 @93=X2.MM32
type=p_mosfet: @104=X3.MM41 @109=X3.MM42 @15=X0.MM1 @25=X0.MM2 @40=X0.MM5 @45=X0.MM6
  @78=X1.MM15 @83=X1.MM16 @98=X2.MM34
type=net: LG_Vbiasn LG_Vbiasn2 LG_Vbiasp1 LG_Vbiasp2 Vinn Vinp Voutn Voutp ...
</GRAPH_NODE_IDMAP>
<PINMAP> ... </PINMAP>
<EDGES> ... </EDGES>

Design bounds:
For tech 65nm, length l must stay within [45nm, 100um] and width w must stay
within [45nm, 500um]. Designs outside this range are invalid.
Fixed global: vdd=1.2V (user-set, not tunable).

Allowed valid target handles and parameter tokens:
  @10: l_MM30, w_MM30      (MM0 -- NMOS input pair, matched with @30)
  @15: l_MM12, w_MM12      (MM1 -- PMOS current source, matched with @25)
  @16: l_MM910, w_MM910    (MM10 -- NMOS current source, matched with @60)
  @35: l_MM4, w_MM4        (MM4 -- NMOS tail)
  @40: l_MM56, w_MM56      (MM5 -- PMOS cascode, matched with @45)
  @50: l_MM78, w_MM78      (MM7 -- NMOS cascode, matched with @55)
  @70: iref                (I0 -- reference current)
  @73, @78, @83: MM14/MM15/MM16   (X1: LG_nmos_l1 -> Vbiasn, Vbiasp1)
  @88, @93, @98: MM31/MM32/MM34   (X2: LG_load_biasn_LV -> Vbiasn2)
  @104, @109, @114: MM41/MM42/MM44 (X3: LG_load_biasp_LV -> Vbiasp2)
```

Matched devices share width and length parameters. For example, `@10` and `@30` share `w_MM30` and `l_MM30`, so each update to these parameters changes both input devices.

Turn 1 User Input Excerpt

```
Your goal is: Achieve gain >= 61.7 dB, power <= 425.7 uW,
-3dB bandwidth >= 43.1 kHz,
30.0 deg <= phase_margin <= 90.0 deg
(each spec has 5% tolerance). Prioritize gain first.
[...]

Your current observation is:
<PERFORMANCE>
<GainAt1kHz>25.0 dB</GainAt1kHz>
<Power>14349 uW</Power>
<Bandwidth3dB>111 kHz</Bandwidth3dB>
<PhaseMargin>-36 deg</PhaseMargin>
<InBandFlatness>0.0 dB (limit 6)</InBandFlatness>
</PERFORMANCE>
<FEATURES>
@10 id=X0.MM0 type=n_mosfet w=w_MM30=34.78u l=l_MM30=166.8n Vgs=496 mV
    Vds=698 mV Vdsat=83.9 mV Ids=438 uA Gm=7 mS region=sat
@15 id=X0.MM1 type=p_mosfet w=w_MM12=17.5u l=l_MM12=69.58n Vgs=-476 mV
    Vds=-286 mV Vdsat=-157 mV Ids=-436 uA Gm=4.6 mS region=sat
@16 id=X0.MM10 type=n_mosfet w=w_MM910=175.1u l=l_MM910=6.297u Vgs=834 mV
    Vds=5 mV Vdsat=367 mV Ids=33.6 uA Gm=66.1 uS region=triode
@35 id=X0.MM4 type=n_mosfet w=w_MM4=4.998u l=l_MM4=199.9n Vgs=834 mV
    Vds=216 mV Vdsat=315 mV Ids=872 uA Gm=2.2 mS region=triode
@40 id=X0.MM5 type=p_mosfet w=w_MM56=117.9n l=l_MM56=20u Vgs=-914 mV
    Vds=-202 mV Vdsat=-471 mV Ids=-22.0 nA Gm=48.2 nS region=triode
@50 id=X0.MM7 type=n_mosfet w=w_MM78=75.97n l=l_MM78=20u Vgs=502 mV
    Vds=706 mV Vdsat=114 mV Ids=11.9 nA Gm=167 nS region=sat
@70 id=X1.I0 type=current_source dc=iref=23u ...
</FEATURES>
```

Turn 1 Original LLM Output Excerpt

```
<think>
The current operating point is far from specs: DC gain is only 25 dB (needs >=61.7 dB),
power is 14.35 mW (needs <=425.7 uW), and phase margin is -36 deg (needs 30-90 deg).
Bandwidth is 111 kHz, which is acceptable, but the main problems are: 1. Excessive bias
current: iref=23u causes excessive DC current. 2. PMOS current source MM1/MM2 is too
short (69.58nm) and wide, allowing too much DC current. 3. NMOS current source MM9/MM10
is very wide (175.1u) and acts as resistor rather than current source; needs moderate
W/L. 4. Input pair MM3/MM0 is too wide and short; needs stronger gm with better ro.
5. Tail MM4 is over-sized in current. 6. PMOS current source MM5/MM6 and NMOS cascode
   MM7/MM8 are weak (only 117.9n/75.97n). 7. DC gain is low because ro of signal devices
   are too small. Strategy: Reduce iref, make PMOS load stronger but less conductive,
   improve input pair, moderate tail, strengthen cascode current source, and fix bias
   blocks. ...
</think>
<action>decrease @70 iref 95%</action>
<action>increase @15 l_MM12 200%</action>
<action>decrease @15 w_MM12 50%</action>
<action>decrease @30 w_MM30 70%</action>
<action>increase @30 l_MM30 200%</action>
<action>decrease @35 w_MM4 50%</action>
<action>increase @35 l_MM4 20%</action>
<action>decrease @16 w_MM910 95%</action>
<action>increase @16 l_MM910 50%</action>
<action>increase @40 w_MM56 500%</action>
<action>decrease @40 l_MM56 99%</action>
<action>increase @50 w_MM78 500%</action>
<action>decrease @50 l_MM78 99%</action>
<action>decrease @73 w_MM14 95%</action>
<action>decrease @73 l_MM14 85%</action>
<action>decrease @78 w_MM15 98%</action>
<action>increase @78 l_MM15 300%</action>
<action>decrease @83 w_MM16 95%</action>
<action>decrease @83 l_MM16 70%</action>
<action>increase @104 w_MM41 500%</action>
<action>increase @109 w_MM42 500%</action>
<action>increase @114 w_MM44 10%</action>
```

Simulator Result After Turn 1

Gain / power	-21.9 dB / 23 μW
Bandwidth	Not measurable
Phase margin	Unavailable

The 22 simultaneous updates produce an invalid design with non-positive gain. The policy receives the simulator measurements and the `design_error` message before the next turn.

Turn 2 User Input Excerpt

```
<PERFORMANCE>
<Status>INVALID - design_error</Status>
<Error>Design produces non-positive gain: -21.86 dB.</Error>
<GainAt1kHz>-21.9 dB</GainAt1kHz>
<Power>23 uW</Power>
<Bandwidth3dB>not measurable</Bandwidth3dB>
<Validity>non-positive gain; power/bandwidth/PM not meaningful until gain > 0</Validity>
</PERFORMANCE>
<FEATURES>
@10 id=X0.MM0 type=n_mosfet w=w_MM30=10.4u l=l_MM30=500.4n
@15 id=X0.MM1 type=p_mosfet w=w_MM12=8.8u l=l_MM12=208.7n
@16 id=X0.MM10 type=n_mosfet w=w_MM910=8.8u l=l_MM910=9.4u
@35 id=X0.MM4 type=n_mosfet w=w_MM4=2.5u l=l_MM4=239.9n
@40 id=X0.MM5 type=p_mosfet w=w_MM56=707.4n l=l_MM56=200n
@50 id=X0.MM7 type=n_mosfet w=w_MM78=455.8n l=l_MM78=200n
@70 id=X1.I0 type=current_source dc=iref=1.2u ...
</FEATURES>
```

Turn 2 Original LLM Output Excerpt

```
<think>
The current operating point is invalid: DC gain is -21.9 dB (magnitude only 0.219x). The
previous aggressive changes destroyed the DC bias: the PMOS current source/load
(MM1/MM2) is too strong, tail MM4 is weak, input pair MM3/MM0 is weak, and NMOS current
source MM9/MM10 is overkill. The core needs to be restored: make PMOS load weaker
(larger l_MM12, smaller MM1/MM2), stronger input pair MM3/MM0, stronger tail MM4,
moderate iref, and adjust NMOS cascode/load. Need positive DC gain first.
</think>
<action>increase @15 l_MM12 400.0%</action>
<action>decrease @15 w_MM12 30.0%</action>
<action>increase @10 w_MM30 100.0%</action>
<action>increase @10 l_MM30 100.0%</action>
<action>increase @35 w_MM4 300.0%</action>
<action>increase @35 l_MM4 50.0%</action>
<action>decrease @16 w_MM910 70.0%</action>
<action>increase @16 l_MM910 20.0%</action>
<action>increase @40 w_MM56 300.0%</action>
<action>increase @40 l_MM56 20.0%</action>
<action>increase @50 w_MM78 200.0%</action>
<action>increase @50 l_MM78 20.0%</action>
<action>decrease @70 iref 10.0%</action>
<action>decrease @73 w_MM14 70.0%</action>
<action>decrease @73 l_MM14 20.0%</action>
<action>increase @78 w_MM15 100.0%</action>
<action>decrease @78 l_MM15 10.0%</action>
<action>decrease @83 w_MM16 70.0%</action>
<action>increase @83 l_MM16 10.0%</action>
<action>increase @98 w_MM34 150.0%</action>
<action>decrease @93 w_MM32 85.0%</action>
<action>decrease @88 w_MM31 70.0%</action>
<action>decrease @88 l_MM31 60.0%</action>
```

Simulator Result After Turn 2

Gain / power	75.6 dB / 98 μW
Bandwidth / phase margin	15 kHz / 42°
In-band peaking	0.0 dB

The design becomes valid and meets the gain, power, and phase-margin targets. Bandwidth remains below its 43.1 kHz target, while gain now has substantial headroom.

Turn 3 User Input Excerpt

```
<PERFORMANCE>
<GainAt1kHz>75.6 dB</GainAt1kHz>
<Power>98 uW</Power>
<Bandwidth3dB>15 kHz</Bandwidth3dB>
<PhaseMargin>42 deg</PhaseMargin>
<InBandFlatness>0.0 dB (limit 6)</InBandFlatness>
</PERFORMANCE>
<FEATURES>
@10 id=X0.MM0 type=n_mosfet w=w_MM30=20.8u l=l_MM30=1u Vgs=324 mV
    Vds=434 mV Vdsat=48.5 mV Ids=3 uA Gm=63.6 uS region=subthreshold
@15 id=X0.MM1 type=p_mosfet w=w_MM12=6.2u l=l_MM12=1u Vgs=-477 mV
    Vds=-760 mV Vdsat=-139 mV Ids=-3.3 uA Gm=42.6 uS region=sat
@16 id=X0.MM10 type=n_mosfet w=w_MM910=2.6u l=l_MM910=11.3u Vgs=446 mV
    Vds=281 mV Vdsat=83.1 mV Ids=351 nA Gm=6 uS region=sat
@35 id=X0.MM4 type=n_mosfet w=w_MM4=10u l=l_MM4=359.8n Vgs=446 mV
    Vds=6.2 mV Vdsat=82.0 mV Ids=6 uA Gm=71.8 uS region=triode
@40 id=X0.MM5 type=p_mosfet w=w_MM56=2.8u l=l_MM56=240n Vgs=-440 mV
    Vds=-110 mV Vdsat=-61.9 mV Ids=-342 nA Gm=7.5 uS region=subthreshold
@50 id=X0.MM7 type=n_mosfet w=w_MM78=1.4u l=l_MM78=240n Vgs=345 mV
    Vds=49.2 mV Vdsat=45.8 mV Ids=341 nA Gm=7.7 uS region=cutoff
@70 id=X1.I0 type=current_source dc=iref=1.1u ...
</FEATURES>
```

Turn 3 Original LLM Output Excerpt

```
<think>
Current specs: Gain is 75.6 dB, well above the 61.7 dB target. Power is only 98 uW, well
within 425.7 uW. Phase margin is 42 deg, which meets 30-90 deg. The unmet spec is
bandwidth: only 15 kHz, needs >=43.1 kHz. The main bottleneck is the high output
resistance: MM3/MM0 are long (1u), MM1/MM2 PMOS load is long (1u), MM9/MM10 current
source is very long (11.3u), and PMOS/NMOS cascode/load devices MM5/MM6 and MM7/MM8 are
modest. The DC gain is excessive because ro of input pair and PMOS load are large. Need
to reduce device lengths, especially ro of signal path devices ... Goal is to lower ro
to reduce Rout, improving -3dB bandwidth while maintaining adequate DC gain.
</think>
<action>decrease @10 l_MM30 70%</action>
<action>increase @10 w_MM30 10%</action>
<action>decrease @15 l_MM12 60%</action>
<action>increase @15 w_MM12 20%</action>
<action>decrease @16 l_MM910 80%</action>
<action>increase @16 w_MM910 50%</action>
<action>increase @40 w_MM56 100%</action>
<action>decrease @40 l_MM56 20%</action>
<action>increase @50 w_MM78 100%</action>
<action>decrease @50 l_MM78 15%</action>
<action>decrease @70 iref 10%</action>
```

Simulator Result After Turn 3

Gain / power	75.9 dB / 169 μW
Bandwidth / phase margin	37 kHz / 52°

Bandwidth increases from 15 to 37 kHz following the combined length, width, and bias updates. It remains below the target even with the 5% evaluation tolerance. The other three targets remain met.

Turn 4 Observation Summary

```
(Observation: <GainAt1kHz>75.9 dB</GainAt1kHz><Power>169 uW</Power><Bandwidth3dB>37 kHz
 </Bandwidth3dB><PhaseMargin>52 deg</PhaseMargin>; input pair w=22.9u l=300n Ids=7.3 uA
 Gm=160 uS; MM1 w=7.4u l=400n; MM10 w=3.9u l=2.3u; MM5 w=5.6u l=192n; MM7 w=2.8u l=204n;
 iref=990n)
```

Turn 4 Original LLM Output Excerpt

```
<think>
Current specs: Gain is 75.9 dB (well above 61.7 dB target), Power is only 169 uW (well
within 425.7 uW), Phase Margin is 52 deg (meets 30-90 deg). The only unmet spec is
Bandwidth: 37 kHz requires >=43.1 kHz. The main bottleneck is the DC gain, dominated by
ro of signal path devices. Need to reduce ro of MM0/MM3 (input pair), MM1/MM2 PMOS load,
MM9/MM10 current source, MM5/MM6 PMOS cascode, MM7/MM8 NMOS cascode. Strategy: decrease
device lengths, especially l_MM30, l_MM12, l_MM910, l_MM56, l_MM78. Also improve iref
slightly for more current.
</think>
<action>decrease @10 l_MM30 50.0%</action>
<action>decrease @15 l_MM12 40.0%</action>
<action>increase @15 w_MM12 10.0%</action>
<action>decrease @16 l_MM910 55.0%</action>
<action>increase @16 w_MM910 15.0%</action>
<action>decrease @40 l_MM56 25.0%</action>
<action>increase @40 w_MM56 10.0%</action>
<action>decrease @50 l_MM78 30.0%</action>
<action>increase @50 w_MM78 15.0%</action>
<action>increase @70 iref 15.0%</action>
```

Final Simulator Result After Turn 4

Gain / power	68.4 dB / 318 μW
Bandwidth / phase margin	172 kHz / 56.4°
In-band peaking	0.0 dB

All four original targets are satisfied without relying on the evaluation tolerance. The episode terminates after four turns.

1.2 Case 2: Out-of-Distribution Current-Mirror Amplifier

Figure 2 shows the current-mirror amplifier used in this case. The second task starts with a gain of -123.2 dB and 61.2 dB of in-band peaking. The input pair and PMOS output mirror are in triode, and bandwidth is not measurable. After two turns, the policy reaches a valid operating point with sufficient gain and bandwidth. Two further updates reduce power and improve phase margin while retaining enough gain to meet the target.

Task Setup	
Topology	Fully-differential current-mirror amplifier
Evaluation split	Out-of-distribution
Technology	65 nm CMOS (PTM), $V_{DD} = 1.2$ V
Gain target	≥ 26.4 dB
Power target	$\leq 170.1\ \mu$W
Bandwidth target	≥ 1.25 MHz (-3 dB bandwidth)
Phase-margin target	$30^\circ \leq \text{PM} \leq 90^\circ$
Evaluation tolerance	5%, as defined in the main evaluation
Initial gain	-123.2 dB
Initial power	$52\ \mu$W
Initial bandwidth	Not measurable
Initial phase margin	Unavailable
Initial in-band peaking	61.2 dB
Initial validity	Invalid under the domain-validity gates.

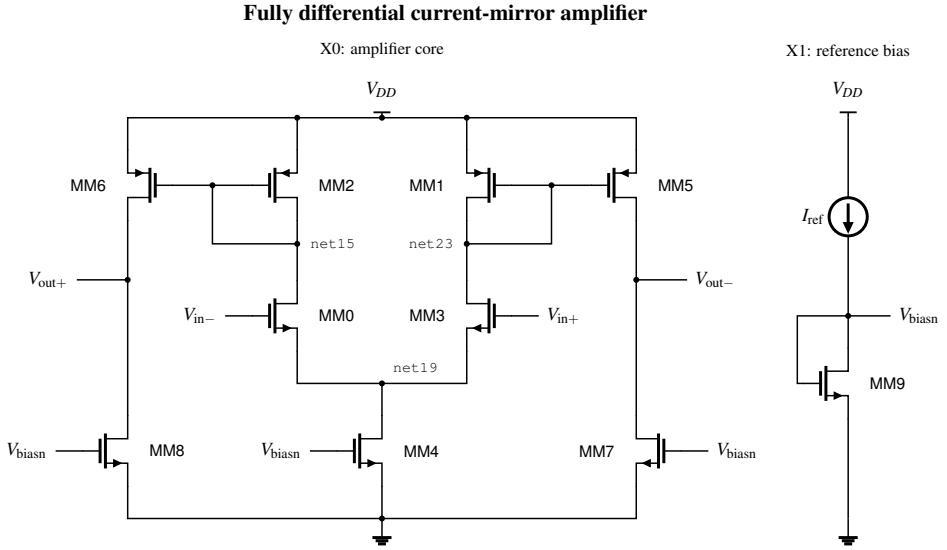


Figure 2: Fully differential current-mirror amplifier used in Case 2. X0 contains the amplifier and X1 supplies its shared bias voltage. Device identifiers and connections follow the supplied netlist. Identical bias labels denote the same electrical net. In both schematics, V_{in+}/V_{in-} and V_{out+}/V_{out-} correspond to `Vinp`/`Vinn` and `Voutp`/`Voutn`, respectively.

Structured System Input Excerpt

```
<GRAPH_INFO>
TOPOLOGY_NAME: Fully_differential_current_mirror
</GRAPH_INFO>
<GRAPH_NODE_IDMAP>
type=pin: @0=vdd @1=vss
type=current_source: @55=X1.I0
type=n_mosfet: @22=X0.MM3 @27=X0.MM4 @42=X0.MM7 @47=X0.MM8 @58=X1.MM9 @7=X0.MM0
type=p_mosfet: @12=X0.MM1 @17=X0.MM2 @32=X0.MM5 @37=X0.MM6
type=net: @2=LG_Vbiasn @3=Vinn @4=Vinp @5=Voutn @6=Voutp ...
</GRAPH_NODE_IDMAP>
<PINMAP> ... </PINMAP>
<EDGES> ... </EDGES>

Design bounds:
For tech 65nm, length l must stay within [45nm, 100um] and width w must stay
within [45nm, 500um]. Designs outside this range are invalid.
Fixed global: vdd=1.2V (user-set, not tunable).

Allowed valid target handles and parameter tokens:
  @7: l_MM30, w_MM30      (MM0 -- NMOS input pair, matched with @22)
  @12: l_MM21, w_MM21     (MM1 -- PMOS diode load, matched with @17)
  @27: l_MM4, w_MM4       (MM4 -- NMOS tail)
  @32: l_MM56, w_MM56     (MM5 -- PMOS output mirror, matched with @37)
  @42: l_MM78, w_MM78     (MM7 -- NMOS output sink, matched with @47)
  @55: iref               (I0 -- bias current source)
  @58: l_MM9, w_MM9       (MM9 -- NMOS bias diode)
```

Turn 1 User Input Excerpt

```
Your goal is: Achieve gain >= 26.4 dB, power <= 170.1 uW,
-3dB bandwidth >= 1.25 MHz,
30.0 deg <= phase_margin <= 90.0 deg
(each spec has 5% tolerance). Prioritize gain first.
[...]

Your current observation is:
<PERFORMANCE>
<GainAt1kHz>-123.2 dB</GainAt1kHz>
<Power>52 uW</Power>
<Bandwidth3dB>not measurable</Bandwidth3dB>
<InBandFlatness>61.2 dB (limit 6)</InBandFlatness>
<Validity>non-positive gain; power/bandwidth/PM not meaningful until gain >
  0</Validity>
</PERFORMANCE>
<FEATURES>
@7 id=X0.MM0 type=n_mosfet w=w_MM30=462.5n l=l_MM30=185.9n Vgs=647 mV
    Vds=970 uV Vdsat=132 mV Ids=188 nA Gm=1.3 uS region=triode
@12 id=X0.MM1 type=p_mosfet w=w_MM21=1.6u l=l_MM21=20u Vgs=-646 mV
    Vds=-646 mV Vdsat=-276 mV Ids=-188 nA Gm=1.2 uS region=sat
@27 id=X0.MM4 type=n_mosfet w=w_MM4=1.8u l=l_MM4=20u Vgs=522 mV
    Vds=553 mV Vdsat=127 mV Ids=428 nA Gm=5.5 uS region=sat
@32 id=X0.MM5 type=p_mosfet w=w_MM56=22u l=l_MM56=417.4n Vgs=-646 mV
    Vds=-32.0 uV Vdsat=-277 mV Ids=-30.5 nA Gm=102 nS region=triode
@42 id=X0.MM7 type=n_mosfet w=w_MM78=58.7n l=l_MM78=7.9u Vgs=522 mV
    Vds=1.2 V Vdsat=127 mV Ids=29.9 nA Gm=381 nS region=sat
@55 id=X1.I0 type=current_source dc=iref=8.2u
@58 id=X1.MM9 type=n_mosfet w=w_MM9=196.1n l=l_MM9=140.5n Vgs=522 mV
    Vds=522 mV Vdsat=118 mV Ids=8.1 uA Gm=93.6 uS region=sat
</FEATURES>
```

Turn 1 Original LLM Output Excerpt

```
<think>
The current operating point is invalid: DC gain is -123.2 dB (magnitude only
0.123 dB), power 52 uW is fine but bandwidth is not measurable because the
low-frequency gain is essentially zero. The main problems are:
1. PMOS load MM1/MM2 is very weak: W/L only 1.6u/20u = 0.08. Needs stronger PMOS
   current source.
2. NMOS pair MM3 is too weak: W/L only 462.5n/185.9n = 2.49, needs wider input
   pair.
3. Tail MM4 is poorly sized: W/L only 1.8u/20u = 0.09, should provide more tail
   current.
4. PMOS current source MM5/MM6 is too wide/triode: 22u/417n acts as resistor
   rather than DC current source.
5. NMOS load MM7/MM8 is too weak: W=58.7n, L=7.9u gives poor DC level shifting.
   Need stronger NMOS load.
6. iref can be reduced later, but first establish DC bias. Strategy: Make MM3
   strong pair, MM1/MM2 moderate PMOS load, stronger tail, better PMOS current
   source, stronger NMOS load, moderate iref.
</think>
<action>increase @22 w_MM30 500%</action>
<action>decrease @22 l_MM30 20%</action>
<action>increase @12 w_MM21 500%</action>
<action>decrease @12 l_MM21 99%</action>
<action>increase @27 w_MM4 500%</action>
<action>decrease @27 l_MM4 99%</action>
<action>decrease @32 w_MM56 95%</action>
<action>increase @32 l_MM56 100%</action>
<action>increase @42 w_MM78 500%</action>
<action>decrease @42 l_MM78 99%</action>
<action>decrease @55 iref 50%</action>
<action>increase @58 w_MM9 100%</action>
<action>increase @58 l_MM9 50%</action>
```

Simulator Result After Turn 1

Gain / power	-118.4 dB / 25 μW
Bandwidth / phase margin	Not measurable
In-band peaking	39.5 dB

The first update does not restore valid operation. Gain remains non-positive and in-band peaking exceeds the validity threshold. The simulator again returns a `design_error`.

Turn 2 Observation Summary

```
(observation: INVALID - design_error, gain -118.4 dB, power 25 uW,
 in-band flatness 39.5 dB; sizes after Turn 1: MM0/MM3 2.8u/148.7n, MM1/MM2
   9.6u/200n,
 MM4 10.8u/200n, MM5/MM6 1.1u/834.8n, MM7/MM8 352.2n/79n, MM9 392.2n/210.8n,
   iref 4.1u)
```

Turn 2 Original LLM Output Excerpt

```
<think>
The current operating point is invalid: DC gain is -118.4 dB (magnitude only
0.118 V/V). The DC gain polarity and magnitude are wrong. The main problems are
the bias: PMOS load MM1/MM2 is too strong compared to NMOS pair MM0/MM3, tail
MM4 current is moderate, PMOS current source MM5/MM6 and NMOS current mirror
MM7/MM8 are not properly sized. The DC operating point is such that Vout is
wrong. Need to restore DC: make NMOS pair MM0/MM3 stronger (larger W, longer L),
PMOS load MM1/MM2 weaker (smaller W, longer L), stronger tail MM4, stronger PMOS
current source MM5/MM6, and better NMOS current mirror MM7/MM8. iref can be
reduced later for power.
</think>
<action>increase @22 w_MM30 500.0%</action>
<action>increase @22 l_MM30 200.0%</action>
<action>decrease @12 w_MM21 80.0%</action>
<action>increase @12 l_MM21 400.0%</action>
<action>increase @27 w_MM4 200.0%</action>
<action>decrease @27 l_MM4 10.0%</action>
<action>increase @32 w_MM56 300.0%</action>
<action>decrease @32 l_MM56 50.0%</action>
<action>increase @42 w_MM78 500.0%</action>
<action>increase @42 l_MM78 100.0%</action>
<action>decrease @55 iref 20.0%</action>
<action>decrease @58 w_MM9 20.0%</action>
<action>decrease @58 l_MM9 20.0%</action>
```

Simulator Result After Turn 2

Gain / power	40.2 dB / 224 μW
Bandwidth / phase margin	6.9 MHz / 8.5°
In-band peaking	0.0 dB

The design becomes valid and exceeds the gain and bandwidth targets. Power remains above the budget, and phase margin is below the accepted range.

Turn 3 Observation Summary

```
(observation: gain 40.2 dB, power 224 uW, bandwidth 7 MHz, PM 8 deg; MM0/MM3
  16.8u/446.1n
 Ids=4 uA, MM5/MM6 4.4u/417.4n Ids=-24.2 uA, MM7/MM8 2.1u/158n Ids=24.2 uA, iref
   3.3u)
```

Turn 3 Original LLM Output Excerpt

```
<think>
Current specs: Gain is 40.2 dB (well above 26.4 dB target), BW is 7 MHz (meets
1.25 MHz requirement), but Power is 224 uW (exceeds 170.1 uW) and Phase Margin
is only 8 deg (needs 30-90 deg). The main issues are excessive bias current and
over-sized PMOS current source/load devices causing too much unity-gain
frequency phase lag. Need to reduce iref, scale down MM5/MM6 PMOS current
source, MM7/MM8 NMOS load, and iref/MM9 bias to save power and improve phase
margin while maintaining gain and bandwidth.
</think>
<action>decrease @55 iref 65.0%</action>
<action>decrease @32 w_MM56 60.0%</action>
<action>decrease @32 l_MM56 40.0%</action>
<action>decrease @42 w_MM78 50.0%</action>
<action>decrease @42 l_MM78 20.0%</action>
<action>decrease @27 w_MM4 40.0%</action>
<action>decrease @27 l_MM4 15.0%</action>
<action>decrease @12 w_MM21 20.0%</action>
<action>decrease @12 l_MM21 20.0%</action>
<action>decrease @58 w_MM9 30.0%</action>
<action>increase @58 l_MM9 10.0%</action>
```

Simulator Result After Turn 3

Gain / power	36.2 dB / 90 μW
Bandwidth / phase margin	4.8 MHz / 23.6°

The combined bias and sizing updates bring power within the budget and improve phase margin from 8.5° to 23.6°. Only the phase-margin specification remains unmet.

Turn 4 Observation Summary

```
(observation: gain 36.2 dB, power 90 uW, bandwidth 5 MHz, PM 24 deg; MM0/MM3
  16.8u/446.1n
 Ids=2.1 uA, MM5/MM6 1.8u/250.4n Ids=-9.4 uA, MM7/MM8 1u/126.4n Ids=9.4 uA, iref
   1.2u)
```

Turn 4 Original LLM Output Excerpt

```
<think>
Current specs: Gain is 36.2 dB (well above 26.4 dB target), Power is only 90 uW
(well within 170.1 uW), Bandwidth is 5 MHz (exceeds 1.25 MHz requirement). The
only unmet spec is Phase Margin, which is 24 deg but requires 30.0 deg to 90.0
deg. The main issue is excessive DC gain due to large ro of signal path devices,
causing too much phase lag at unity-gain frequency. Need to reduce ro of
output/load devices to lower DC gain and improve phase margin while maintaining
gain and bandwidth. Strategy: reduce device lengths (MM0/MM3, MM1/MM2, MM5/MM6,
MM7/MM8) and iref to lower ro and DC current, decreasing DC gain and improving
phase margin. Decrease iref moderately to save power.
</think>
<action>decrease @7 w_MM30 50.0%</action>
<action>decrease @7 l_MM30 40.0%</action>
<action>decrease @12 w_MM21 30.0%</action>
<action>decrease @12 l_MM21 40.0%</action>
<action>decrease @32 w_MM56 50.0%</action>
<action>decrease @32 l_MM56 40.0%</action>
<action>decrease @42 w_MM78 40.0%</action>
<action>decrease @42 l_MM78 30.0%</action>
<action>decrease @55 iref 30.0%</action>
<action>decrease @27 w_MM4 30.0%</action>
<action>decrease @27 l_MM4 15.0%</action>
<action>decrease @58 w_MM9 20.0%</action>
<action>decrease @58 l_MM9 10.0%</action>
```

Final Simulator Result After Turn 4

Gain / power	28.4 dB / 78.8 μW
Bandwidth / phase margin	10.2 MHz / 46.8°
In-band peaking	0.0 dB

All four original targets are satisfied without relying on the evaluation tolerance. Gain remains 2.0 dB above the target, while phase margin reaches 46.8° and power remains within the budget. The episode terminates after four turns.

These examples illustrate how simulator feedback can guide recovery from invalid operation and subsequent refinement of coupled performance metrics. The two rollouts demonstrate successful optimization from different initial conditions and specification targets on an in-distribution topology and a held-out topology.